

PS26055

ϵ -Greedy Benchmark & Published-Result Comparison

Evidence dossier for the Electronic Support smart-scan scheduler

Current empirical leader

ϵ -Greedy ($\epsilon=0.10$) reaches **0.06556 hit rate** and **9,834 hits** on the full 250-file TSRD Stare benchmark - **3.31×** the Round Robin baseline.

Benchmark basis

The project report records a 36-band $\times$ 600-slot grid, 500 MHz IBW across an 18 GHz surveillance range, 50 ms slots, a 30 s mission, seed=42, and the PDW-only direct occupancy path. All 250/250 Stare files were admitted; the corpus contains 367,473,884 pulses. The report also records Gate 0 determinism and Gate 2 no-truth-leakage PASS results.

How to read the comparisons

Our result: full-corpus project benchmark. **Published result:** a number explicitly reported by an external paper. **Context:** a useful reference from a different task or metric and therefore not an apples-to-apples performance claim.

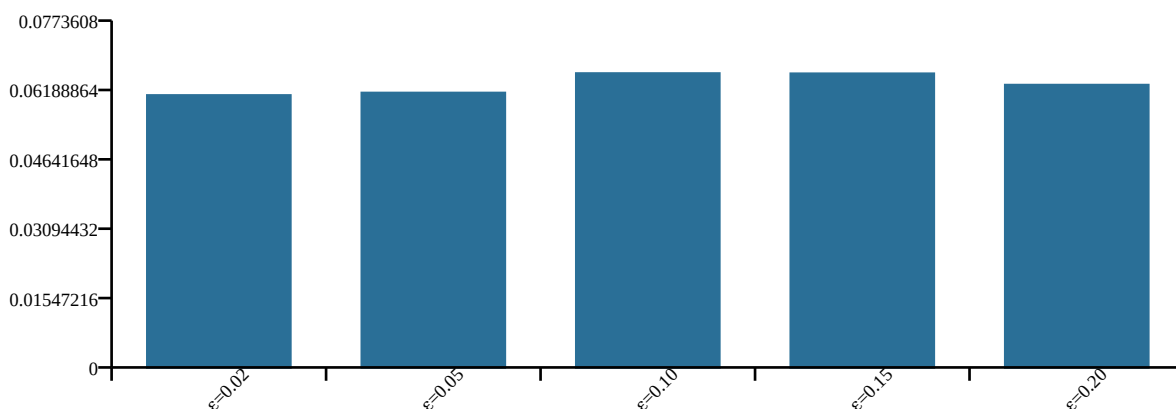
This dossier is intended to become the quantitative benchmark section of the PS26055 book/paper.

1. ϵ -Greedy Benchmark Sweep

The ϵ sweep is the cleanest parameter study currently available for the scheduler. The best configuration is shallow rather than singular: $\epsilon=0.10$ and $\epsilon=0.15$ are effectively tied at the reported precision.

ϵ	Hit rate	Total hits	Relative to RR
0.02	0.06067	9,100	3.06×
0.05	0.06122	9,183	3.09×
0.10	0.06556	9,834	3.31×
0.15	0.06551	9,827	3.31×
0.20	0.06298	9,447	3.18×

ϵ -Greedy sensitivity on the full TSRD corpus



Comparison	$\epsilon=0.10$ relative improvement
vs Round Robin	230.8%
vs UCB1	71.1%
vs Thompson	83.4%
vs SW-UCB w=50	55.2%
vs Weighted Prior p=0.1	24.6%

Interpretation: exploration has an interior optimum. Too little exploration under-explores the corpus; too much exploration spends too much decision budget away from the currently useful bands.

2. Full Scheduler Benchmark Ladder

These are the strongest directly comparable results because every value comes from the same 250-file TSRD benchmark. They form the reference ladder that future DRL and Offline-RL methods must beat.

Scheduler	Hit rate	Hits	vs RR
Round Robin	0.01982	2,973	1.00×
Random Markov (10-run)	0.02552	3,828	1.29×
Clarkson Coprime	0.02619	3,929	1.32×
Thompson Sampling Beta(1,1)	0.03574	5,361	1.80×
UCB1	0.03831	5,747	1.93×
Sliding-window UCB w=50	0.04225	6,338	2.13×
Weighted Prior p=0.1	0.05261	7,892	2.66×
ϵ -Greedy $\epsilon=0.02$	0.06067	9,100	3.06×
ϵ -Greedy $\epsilon=0.05$	0.06122	9,183	3.09×
ϵ -Greedy $\epsilon=0.10$	0.06556	9,834	3.31×
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Key empirical findings

- **ϵ -Greedy $\epsilon=0.10$ is the current overall leader.**
- **Weighted Prior p=0.1** is the strongest listed pure-prior/bandit configuration at 0.05261.
- **Sliding-window UCB w=50** reaches 0.04225 versus UCB1 0.03831, demonstrating that shorter adaptive memory can help under changing dynamics.
- The two Clarkson number-theoretic schedules do not maximize hit rate on this heterogeneous corpus, but that does not invalidate their separate role as synchronization-aware analytical references.

Negative result worth preserving: Hierarchical Thompson failed on 115/250 files because a fixed k=18 clustering assumption did not hold for all emitter configurations. This is direct evidence that extra structural complexity needs a robustness argument.

3. Our Results vs Published Results

The table deliberately avoids false apples-to-apples claims. The literature spans passive ES receiver scheduling, cognitive-radar resource management, and signal-level radar recognition. The metrics differ.

Study / method	Published metric or finding	Reported value	Comparison status
Clarkson et al. (2006)	Intercept-time / MEIT behavior	230.4 ms in cited 25-band example	Context - different metric and analytical model
Qin et al. (2023), LSTM-TD3	CPU parameter-selection time	~10× faster than optimization solver	Context - related cognitive-radar RRM
Lu et al. (2025), SAC vs DDPG	Learning stability / sample efficiency	SAC reported more stable and sample-efficient	Context - multi-objective radar RRM
Lu et al. (2025), SAC study	Pareto-front computation	~48 min for 60 β values; NSGA-II ~11.5 h	Context - offline computational study
Lu et al. (2026), CDRL-DDPG	Decision latency	1.9 ms	Context - continuous dwell allocation
Lu et al. (2026), optimization	Decision latency	5.4 ms	Context - same task; different action space
Lu et al. (2026), CDRL-DDPG	Track initiation time	6 s in 3-of-4 case	Context - tracking metric, not ES interception
Liu et al. (2026), PKM-Net	Accuracy / latency / footprint	99.88% / 1.64 ms / 10.43 MiB	Context - signal-level mode recognition
Sekhar et al. (2025), Double DQN ES	Receiver scheduling robustness	Qualitative robustness result	Direct task precedent; exact absolute metric not verified

Scientific reading of the literature

Clarkson establishes why periodic schedules can suffer synchronization pathologies. Qin demonstrates that memory-based DRL can reduce online parameter-selection cost relative to iterative optimization. Lu et al. show that modern constrained / multi-objective DRL can provide adaptive resource allocation and report concrete millisecond-scale decision latencies. PKM-Net demonstrates that a compact Transformer + GRU can support real-time signal-level inference, but it is not a band scheduler.

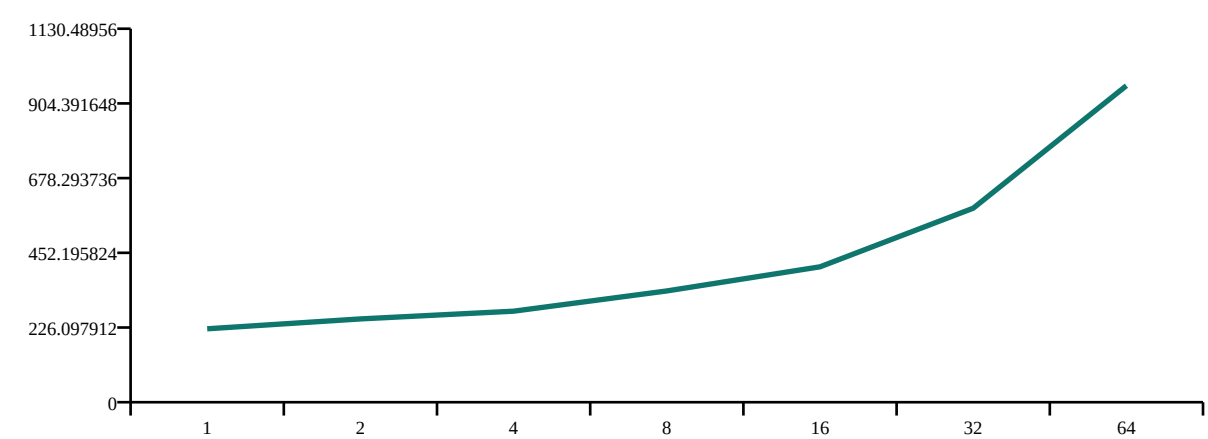
For PS26055, these are supporting arguments - the decisive benchmark remains the common 250-file TSRD experiment.

4. Latency Evidence and the 32-Step Problem

Latency must be reported by percentile and by function. Training time is not the same as inference time; inference time is not the same as RF end-to-end latency. The values below are references with their definitions retained.

System	Latency	What the number means
Our classical scheduler	~18 μ s P50	Earlier software scheduler-loop benchmark; environment/hardware spec
Our 32-history Transformer + GRU	587 μ s P50; 629 μ s P95; 679 μ s P99	History-scaling run on CPU
Detailed 32-history rerun	590 μ s P50; 756 μ s P95; 1.18 ms P99	End-to-end 32-history path incl. tensor construction
Qin et al. 2023 LSTM-TD3 / TD3	~10 $\times$ faster than solver	CPU parameter-selection time
Lu et al. 2026 CDRL-DDPG	1.9 ms	Online decision latency
Lu et al. 2026 optimization	5.4 ms	Per-decision optimization latency
Liu et al. 2026 PKM-Net	1.64 ms	Online signal-level mode recognition

Measured P50 latency vs history length



Finding: moving from 16 to 32 history increases median inference from about 410 μ s to 587 μ s; 64 history approaches 958 μ s. Tensor construction itself was about 1.2 μ s P50 in the detailed rerun, so the dominant cost is repeated neural processing, not making the sliding window.

Design response: initialise long context once, retain GRU hidden state, use shorter local attention or cached attention, and invoke expensive temporal inference selectively when uncertainty/change justifies it.

5. Frozen Benchmark Protocol for DRL / Offline RL

Every new learned scheduler should face the same reference. This prevents improvements caused by different datasets, action spaces, seeds or metrics from being mistaken for algorithmic superiority.

Dimension	Frozen reference
Corpus	250/250 TSRD Stare files; PDW-only direct occupancy path
Grid	36 bands × 600 slots
Timing	50 ms slots; 30 s mission
Bandwidth	18 GHz total; 500 MHz IBW
Seed	42 for the published project benchmark; future work should add multi-seed confidence intervals
Current reference policy	ϵ -Greedy, $\epsilon=0.10$
Primary metric	Hit rate
Secondary metrics	Total hits, coverage, Pd/Pfa where defined, miss rate, intercept rate/time, regret/cost
Latency	P50/P95/P99/P99.9/max; disclose CPU/GPU, threads, batch, precision and sequence length
Integrity	Gate 0 determinism and Gate 2 no-truth-leakage requirements remain mandatory

Decision ladder

Stage	Candidate	Required evidence to advance
B0	Round Robin / Random	Non-learning floor
B1	ϵ -Greedy $\epsilon=0.10$	Current empirical reference: 0.06556
B2	Double DQN	Scheduler-level gain over B1 with acceptable latency
B3	GRU-DDQN / recurrent DRL	Measurable temporal-memory benefit
B4	Discrete SAC	Gain over best prior DRL/bandit candidate
B5	CQL / IQL	Offline-RL gain with heterogeneous logged data and OOD analysis
B6	Decision Transformer / Transformer+GRU	Value of long history must justify latency
B7	Confidence-gated hybrid	Only surviving components are composed

Final benchmark conclusion. The present project evidence is strong enough to freeze ϵ -Greedy $\epsilon=0.10$ as the reference scheduler for the next research phase. The correct scientific question is no longer “which algorithm sounds strongest?” but “which learned policy can beat 0.06556 on the same corpus while preserving robustness, coverage, adaptation and a measured latency budget?”

Sources used: PS26055 Research Progress Report (05 Sep 2026); PS26055 revised architecture and state-of-the-art documents; Clarkson, El-Mahassni & Howard (2006), DOI 10.1049/ip-rsn:20050055; Qin et al. (2023), DOI 10.1049/rsn2.12469; Lu et al. (2025), arXiv:2506.20853; Lu et al. (2026), arXiv:2606.05526 / IEEE T. Radar Systems; Liu et al. (2026), IEEE TNNLS, DOI 10.1109/TNNLS.2026.3682455; Sekhar et al. (2025), MILCOM Double-DQN ES receiver scheduling.